Assessment and extension of chest X ray severity prediction

DSA5206 Type 2 Project Part II

Cohen et al. 2020 | Classroom checklist from Lecture 9 | Updated 27 October 2026

We assess whether the study supports its claims and whether a different model improves prediction of expert radiographic severity scores. The first comparison changed only the regression head. This extension also changes the image representation, using released BioViL, BioViL-T and RAD-DINO encoders. Across the initial and extended analyses, we ran 3,870 outer fits on 94 chest radiographs from 65 patients. All configurations and predictions are retained.

Geographic Extent: BioViL PCA ridge has MAE 0.763 ± 0.073, 31.1% lower than Bounded OLS 1 (1.107).

Opacity: BioViL-T PCA ridge has MAE 0.662 ± 0.058, 11.2% lower than Bounded OLS 4 (0.745).

These are exploratory internal results. The extension was designed after reviewing the first experiment. We therefore show training-only model selection alongside the observed leaderboard, rather than treating the lowest test error as an independently confirmed winner.

Results to read first

Comparison	Extent MAE	Opacity MAE
Original paper method	1.165 ± 0.104	0.766 ± 0.072
Same method with score bounds	1.107 ± 0.110	0.745 ± 0.072
Best observed modern pipeline	0.763 ± 0.073	0.662 ± 0.058
Training selected OLS	1.127 ± 0.107	0.766 ± 0.071
Training selected modern	0.827 ± 0.120	0.701 ± 0.080
Training selected all	0.827 ± 0.120	0.743 ± 0.115

Mean ± SD over 50 overlapping patient splits; lower is better. SD is not a confidence interval. The original-method rows are our reruns, not values copied from the publication. Best observed pipeline names are given in the experimental results.

The targets are geographic extent on a 0-8 scale and opacity on a 0-6 scale. Predicting these scores does not establish diagnostic accuracy, prognosis or benefit to patient care. [1]

Checklist assessment of data collection

Lecture 9 lists the 12 questions on p. 11 and discusses them on pp. 14-21. The judgments below concern the 2020 study. Our new experiments are evaluated separately.

1 Is the data collection approved

Ethics approval and consent handling are reported

Paper evidence. The Human Subjects statement reports approval CERSES-20-058-D and states that consent was obtained or waived. Images were collected from existing public sources and de-identified before analysis. [1, pp. 2 and 7]

Our reading. This supports ethical oversight of the retrospective study. Public availability alone would not establish permission for every secondary use. No prospective study registration is reported; that reporting gap should not be treated as evidence that the study lacked ethics approval.

What would strengthen the evidence. Retain the approval identifier, data attribution and applicable source terms. A future prospective clinical evaluation would require an appropriate protocol and approvals.

Lecture 9, pp. 11 and 14

2 Is the data collected properly

Sources are described but representativeness is uncertain

Paper evidence. The study uses 94 PA chest radiographs from a public collection, sourced internationally from December 2019 to March 2020. Patients were reported COVID-19 positive, mostly by RT-PCR. [1, p. 2, Section 2.1]

Our reading. The collection is a small convenience sample assembled partly from publications and image-sharing platforms, rather than a defined consecutive clinical cohort. International sources provide diversity but do not establish representative sampling or independent external validation. Exclusion rules and the patient flow are insufficiently detailed.

What would strengthen the evidence. Publish a case flow, patient-level denominators, acquisition settings and inclusion criteria. Validate on consecutive cases from hospitals not involved in model development.

Lecture 9, pp. 11 and 14

3 Is the data of good quality

Expert labels are a strength but label uncertainty remains

Paper evidence. Three blinded experts, including two experienced chest radiologists, scored geographic extent and opacity. Fleiss' kappa was 0.71 for extent and 0.45 for opacity. The paper states that there was no missing data. [1, p. 2, Sections 2.1 and 2.2]

Our reading. Expert review provides a defensible reference for radiographic severity, although opacity labels show appreciable reader disagreement. A mean expert score is not a clinical outcome. In our fixed public snapshot all 94 targets are present, but age is missing for 25 image records and sex for 17; the paper's missingness statement should therefore not be generalized to all clinical metadata.

What would strengthen the evidence. Report missingness separately for targets and covariates, distinguish images from patients, inspect duplicates and image artifacts, and account for reader variability when interpreting model error.

Lecture 9, pp. 11 and 15

Checklist assessment of preprocessing and models

Lecture 9 lists the 12 questions on p. 11 and discusses them on pp. 14-21. The judgments below concern the 2020 study. Our new experiments are evaluated separately.

4 How is data preprocessed

The main image transformations are clearly specified

Paper evidence. Images are center-cropped where needed, resized to 224 by 224 pixels and scaled to approximately -1024 to 1024. The pretrained network is frozen before generating features for the regression models. [1, pp. 2-3, Sections 2.4 and 2.5]

Our reading. This gives a reproducible basic pipeline. However, center-cropping can remove peripheral anatomy, and resizing can obscure small findings. The study does not establish robustness to these choices, acquisition artifacts or differences in radiographic views. Frozen feature extraction does not itself require fitting on the COVID-19 test labels.

What would strengthen the evidence. Record implementation versions and test clinically plausible preprocessing changes. Any fitted scaler, PCA or feature selector introduced in an extension must be fitted within training folds only.

Lecture 9, pp. 11 and 15

5 What ML models are used

A sensible transfer learning design with limited alternatives

Paper evidence. DenseNet was pretrained on 88,079 non-COVID-19 images from seven datasets. Frozen representations of 1024, 18, 4 or 1 dimensions feed ordinary linear regression, alongside an intercept-only baseline. The task outputs are pre-sigmoid logits. [1, pp. 2-4 and Figure 1]

Our reading. Learning only a small regression head is appropriate for scarce labels. The reported 1+1 parameter count describes the new regression head, not the whole pretrained network. Comparing unregularized high-dimensional regression with a single feature does not establish that high-dimensional representations contain less useful information.

What would strengthen the evidence. Compare regularized regressors on the same frozen features and patient splits. We test Ridge regression and RBF support vector regression, with training-only hyperparameter selection.

Lecture 9, pp. 11 and 16

6 How is the data split

Patient grouping is appropriate but evaluation is internal

Paper evidence. The 94 images are repeatedly divided approximately 50/50, keeping all timepoints of the same patient together. Table 1 summarizes 50 random train/test splits. [1, pp. 3-4, Section 2.5 and Table 1]

Our reading. Patient grouping addresses direct leakage between repeated scans. The image counts need not be exactly balanced because the allocation is by patients. Repeated test sets overlap, so their results are dependent. Ranking feature configurations on the same repeated test sets also leaves uncertainty about model-selection optimism.

What would strengthen the evidence. Save patient split manifests, reserve test predictions for evaluation and tune alternatives using nested grouped validation. For deployment claims, add hospital-level or temporal external testing.

Lecture 9, pp. 11 and 17

Checklist assessment of evaluation

Lecture 9 lists the 12 questions on p. 11 and discusses them on pp. 14-21. The judgments below concern the 2020 study. Our new experiments are evaluated separately.

7 Are the metrics appropriate

Regression metrics are suitable but uncertainty and calibration need more work

Paper evidence. Table 1 reports MAE, MSE, R-squared and Pearson correlation as means and standard deviations over 50 splits. Figure 2 compares predictions with expert scores and shows systematic overestimation at low scores and underestimation at high scores. [1, p. 4]

Our reading. MAE is interpretable in score units and MSE emphasizes extreme errors. Pearson correlation is not accuracy and can remain high despite miscalibration. Split standard deviations are not confidence intervals. The clinical significance of a one-point error is not established, and subgroup performance and prediction intervals are absent.

What would strengthen the evidence. Use held-out MAE as the primary endpoint, retain complementary metrics and inspect score-stratified signed errors. State the dependence of repeated splits and avoid claiming statistical superiority from a small mean difference alone.

Lecture 9, pp. 11, 17 and 23

8 Are the results explainable

Qualitative explanations are provided but not validated for fidelity

Paper evidence. Figure 4 displays smoothed input-gradient saliency for selected correct and incorrect predictions; Figure 3 visualizes four-dimensional features with t-SNE and survival markers. The authors acknowledge limitations of saliency maps. [1, pp. 3, 5-6]

Our reading. These plots help inspect plausibility, but visually plausible heatmaps do not prove causal reliance on pathology. The t-SNE image is exploratory and does not validate outcome prediction. The public code uses squared gradients with Gaussian sigma 5, whereas the paper describes a 5 by 5 kernel; those are different specifications.

What would strengthen the evidence. Disclose case-selection and smoothing rules, test image-region occlusion and model randomization, and measure sensitivity to text or borders. Avoid interpreting t-SNE clusters as proof of mortality prediction.

Lecture 9, pp. 11 and 18

9 Are the results reproducible and generalizable

Broadly reproducible with versioning gaps; generalizability unproven

Paper evidence. The authors release data, severity labels, code and model resources. The group notebook reproduces the broad Table 1 ranking. The paper calls for new cohorts with the same scoring system. [1, pp. 1 and 6; 4-5]

Our reading. Open resources enable verification, but exact recreation is affected by changing metadata and software. The paper describes 208 visualization images while the supplied author notebook stores 199. The group repository does not include its custom metadata_2020.csv or a pinned environment. None of these facts establishes performance on a genuinely new hospital cohort.

What would strengthen the evidence. Our experiment fixes a public 2020 commit, records checksums, caches features and publishes split manifests and versions. External validation remains a required next step; close internal reproduction does not demonstrate it.

Lecture 9, pp. 11 and 19

Checklist assessment of clinical use

Lecture 9 lists the 12 questions on p. 11 and discusses them on pp. 14-21. The judgments below concern the 2020 study. Our new experiments are evaluated separately.

10 Is there an appropriate clinical context

The use case is relevant but the evaluated endpoint is a surrogate

Paper evidence. The paper proposes assisting serial monitoring and escalation or de-escalation of care, especially in intensive care. Its supervised targets are radiographic extent and opacity. The discussion acknowledges uncertainty about their relationship with functional and clinical outcomes. [1, pp. 1 and 6]

Our reading. Quantitative CXR scoring is a clinically understandable research task. However, agreement with radiologists' scores does not show improved management, reduced mortality or safe triage. Restriction to PA images also limits direct applicability to portable AP examinations in intensive care.

What would strengthen the evidence. Define intended users, acquisition settings and decision points. Evaluate the added value over expert scoring and clinical variables in prospective workflows before claiming clinical benefit.

Lecture 9, pp. 11 and 20

11 What are its limitations

Important limitations are acknowledged and additional ones remain

Paper evidence. The discussion identifies small sample size, sampling bias, difficulty obtaining labels, the need for new cohorts and uncertain linkage to clinical outcomes. Figure 2 illustrates errors at the extremes. [1, pp. 4 and 6]

Our reading. Additional concerns are limited patient numbers, reader disagreement, incomplete demographic reporting, lack of independent external testing, no formal uncertainty estimates for individual predictions and limited alternative regressors. For one-feature models both predicted scores are affine functions of the same scalar and cannot vary independently.

What would strengthen the evidence. Prioritize external validation, patient-level reporting and error analysis by score and acquisition setting. Our alternative-model comparison addresses only one limitation: the narrow regression comparison; it does not resolve the clinical evidence gaps.

Lecture 9, pp. 11 and 21

12 Is the method ready for deployment

No clinical deployment readiness has been demonstrated

Paper evidence. The authors caution against premature implementation and state that diagnostic performance requires appropriate clinical evaluation. The study is a retrospective proof of concept without a prospective workflow or clinical impact trial. [1, p. 6, Discussion]

Our reading. The evidence supports a research tool for approximating expert radiographic scores. It does not establish safe autonomous decisions, stable performance across hospitals and devices, or effective human-AI use. Output values can fall outside the score range, and the implications of severe underestimation have not been resolved.

What would strengthen the evidence. Require independent multicenter and temporal validation, prospective human-AI evaluation, uncertainty and failure handling, monitoring for drift and a defined accountable clinical workflow before routine use.

Lecture 9, pp. 11 and 21

Why and how we expanded the experiment

The initial Ridge and SVR comparison tested whether the same DenseNet features could support a better regression head. It could not test progress in image representation learning. We added three released chest radiograph encoders and retained the old features so that this distinction could be examined.

Encoder	Features used	Scope of this run
DenseNet121	1, 4 or 18 logits; 1024 pooled features	Original frozen transfer learning approach at 224 pixels.
BioViL 2022	2048 pooled; 128 projected	Image encoder from image and report pretraining; official 512 resize and 480 crop. [5]
BioViL T 2023	512 pooled; 128 projected	Single current image only; official 512 resize and 448 crop. Prior scans are not supplied. [6]
RAD DINO 2024 preprint 2025 journal	768 CLS token	Released public image-only self-supervised encoder, with its native image processor. [7,8]

This is a transfer experiment using public checkpoints. We do not reproduce the full pretraining or the separate benchmark tasks in the encoder papers. The released RAD-DINO checkpoint also differs from the paper checkpoint, which used some private data. Revisions, weight hashes, preprocessing and feature arrays are recorded in data/extended_feature_provenance.json. [8]

Regression and preprocessing choices

The 27 new configurations include bounded OLS references; Huber, ElasticNet, kernel ridge, Extra Trees, random forest and gradient boosting on low-dimensional DenseNet outputs; and Ridge, PCA plus Ridge, PLS and SVR on selected high-dimensional representations. The full list and grids were locked before running the expanded comparison. All scalers, variance filters, principal components and regression parameters are fitted within training folds.

New configurations constrain predictions to the known score ranges during both inner validation and test prediction. To separate this simple correction from model gains, we report raw OLS and bounded OLS references. Raw predictions are also retained for every new model. The older unbounded results, including the failed Ridge configuration, remain unchanged.

Evaluation and model selection

Decision	Implementation
Cohort	94 scored PA images and 65 patients reconstructed from the fixed public snapshot of 30 March 2020. This is not a claim of an identical original software environment. [3]
Main comparison	The same 50 patient-grouped half-training half-test splits as the initial experiment, with seeds 0-49.
Parameter choice	Five-fold GroupKFold within each outer training set. Select parameters by mean inner-fold MAE. No outer test patients are used.
Family choice	Training selected OLS chooses between bounded OLS 1 and OLS 4. Training selected modern chooses among 12 modern-encoder pipelines. Training selected all chooses among all 27 new configurations.
Sensitivity analysis	Five outer GroupKFold partitions. Every patient is tested once; pooled error uses all 94 held-out image predictions. Training sets are larger than in the main experiment.
Reported endpoints	MAE, MSE, R squared, correlation, patient-weighted MAE, signed error, high-score error and out-of-range raw predictions.

Choosing the main result without hiding selection

The observed best pipeline is useful for deciding what to investigate next, but its test error benefits from comparing many candidates. We mark that ranking as exploratory. The three training-selected procedures answer a different question: how well would an investigator perform if the model family had to be chosen without seeing the outer test results? Their held-out errors include the cost of selecting an unsuitable family on a small training set. [9]

The extension follows the initial analysis and uses the same cohort. Nested selection prevents direct test-label use within a run, but cannot turn this adaptive study into a prospectively untouched evaluation. The additional five-fold run checks sensitivity to training size and partitioning; it is not an external validation cohort.

Image-weighted MAE remains the primary metric for comparison with the study. We also report patient-weighted error because some patients contribute more than one image. Resampled partitions overlap, so split win rates are descriptive. Conditional bootstrap intervals use the stored patient errors and do not include uncertainty from retraining.

Main results from the expanded comparison

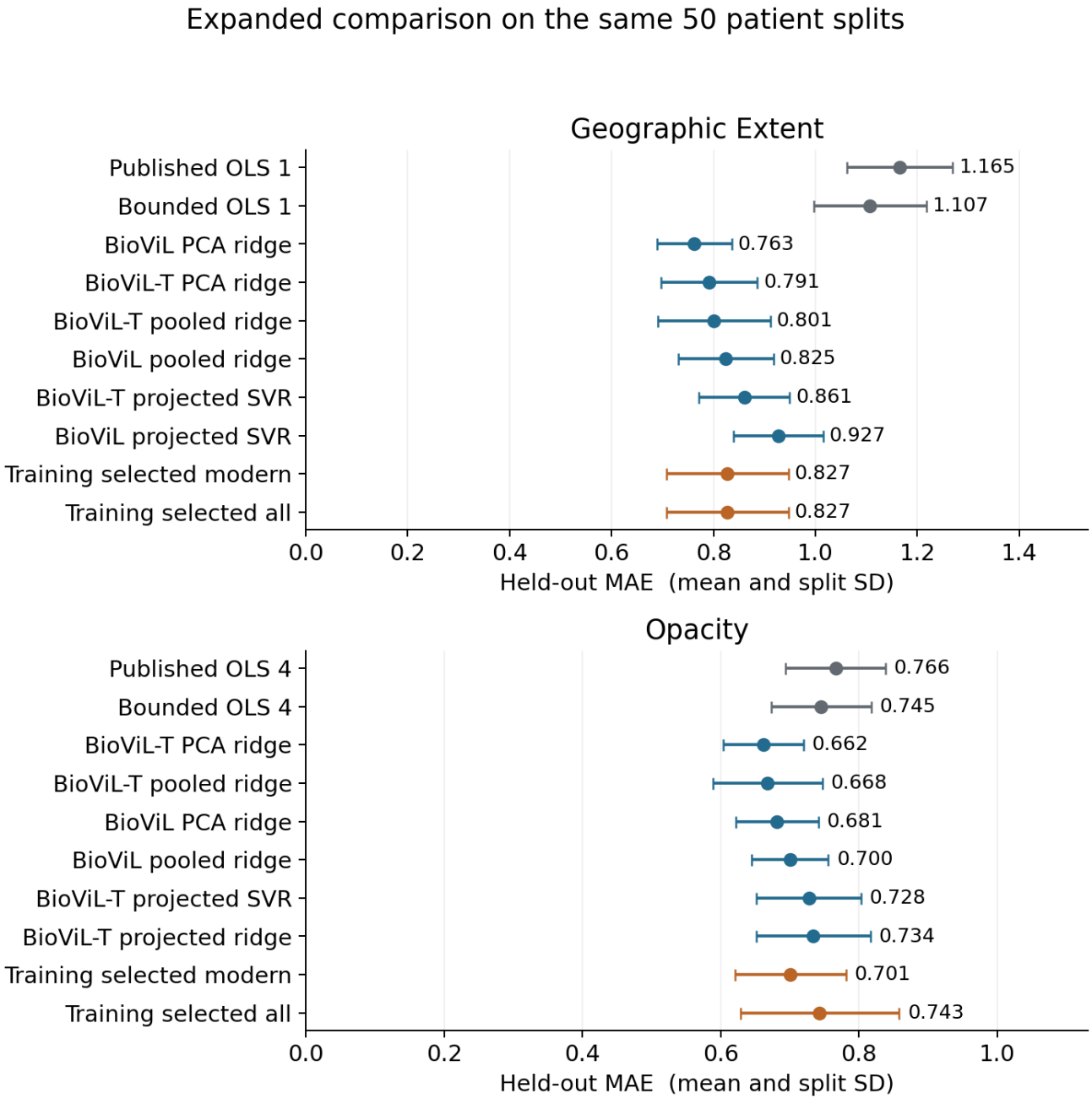


Figure 1. Gray rows are paper-method references, blue rows are the six lowest observed alternative MAEs for each target, and orange rows are training-selected procedures. Error bars are split SD. Ranking the blue rows is exploratory.

Geographic Extent: BioViL PCA ridge has MAE 0.763 ± 0.073, 31.1% lower than Bounded OLS 1 (1.107).

Training selected modern changes MAE by -0.280 against the bounded reference and has lower MAE in 96% of paired splits. These split comparisons reuse patients and should not be read as 50 independent trials.

Opacity: BioViL-T PCA ridge has MAE 0.662 ± 0.058, 11.2% lower than Bounded OLS 4 (0.745).

Training selected modern changes MAE by -0.044 against the bounded reference and has lower MAE in 68% of paired splits. These split comparisons reuse patients and should not be read as 50 independent trials.

The complete leaderboard includes all 36 actual model configurations from both stages, plus three selection procedures. We retain poor results so that the main figures do not imply that every attempted model improved performance.

How much depends on model choice

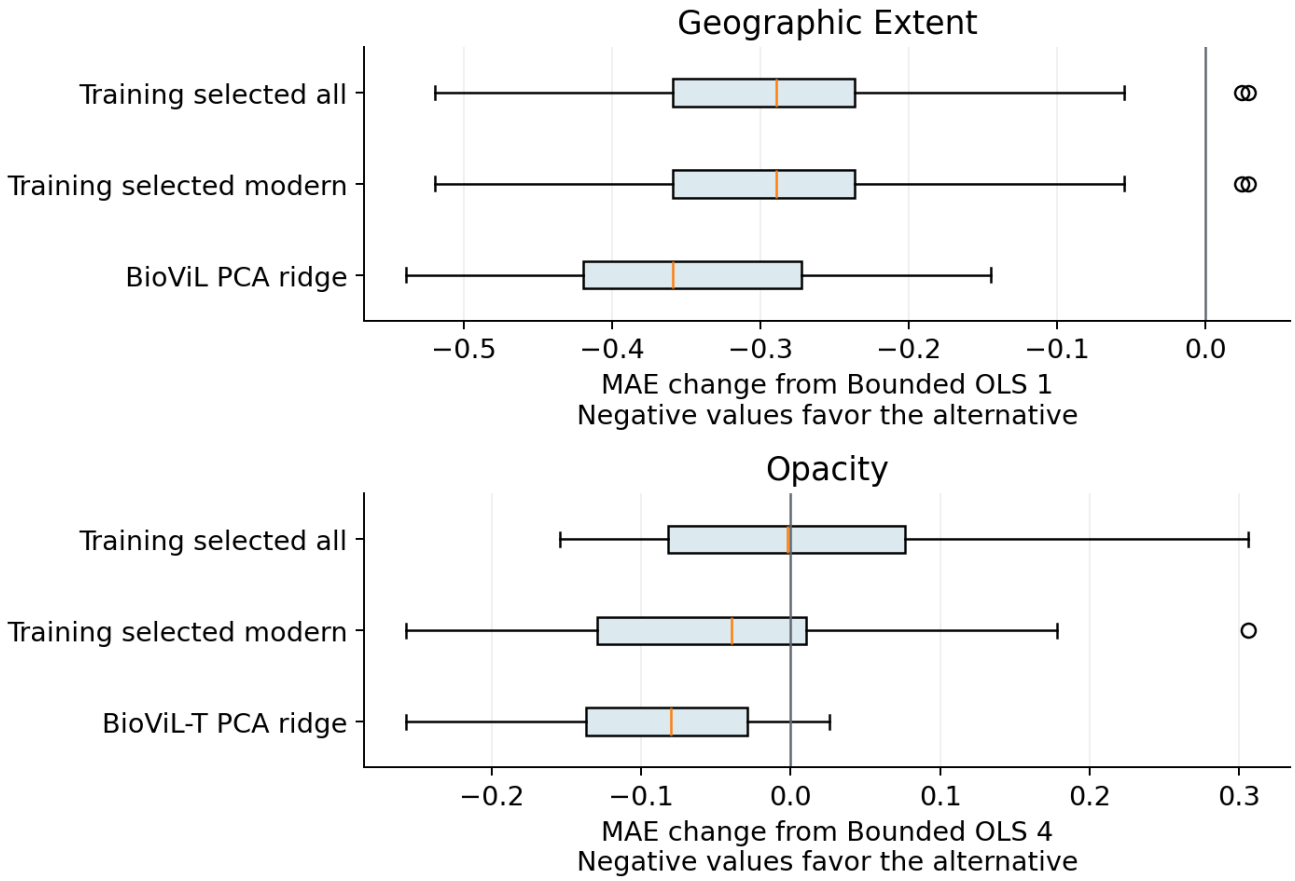


Figure 2. Paired differences from the bounded original reference on each of the 50 test partitions. Negative values favor the alternative. Boxes describe dependent resampling results and are not significance tests.

There is a practical distinction between a pipeline with a low average test error and a reliable rule for finding that pipeline. With roughly 32 training patients in the main experiment, inner validation can rank families differently after a small change of patients. We therefore use the training-selected results to qualify any claim drawn from the observed winner.

For opacity, selecting among all 27 new configurations gives MAE 0.743, compared with 0.701 when selection is restricted to the 12 modern-encoder pipelines. The wider search is almost tied with bounded OLS 4 at 0.745. Adding candidates did not improve the rule for choosing a model in this sample. This is consistent with noisy inner rankings, rather than a reason to hide the weaker selection result.

Which models the training folds selected

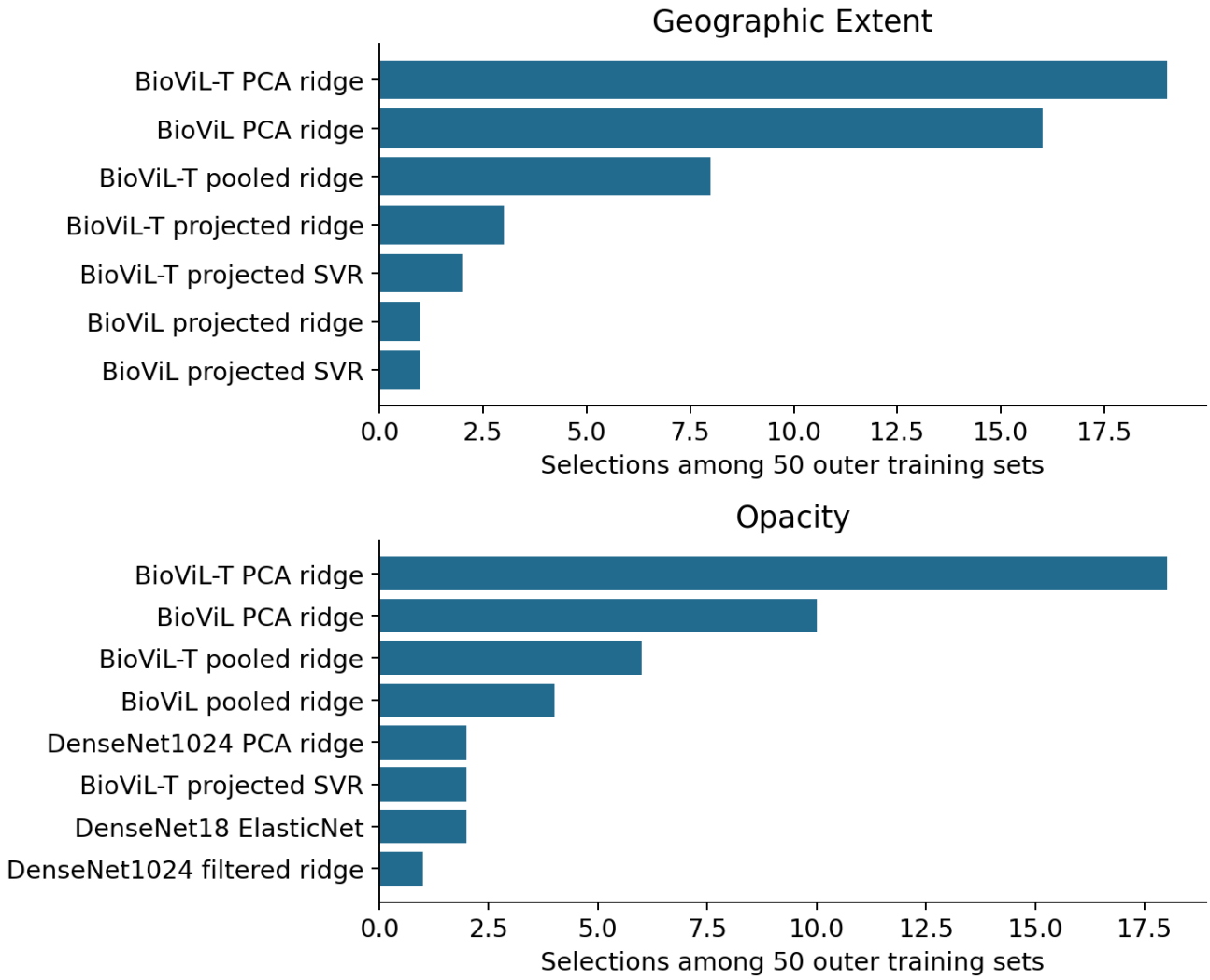


Figure 3. The most frequent model choices under Training selected all, based only on each outer training set. A changing winner indicates uncertainty in selecting a single pipeline from this cohort.

For extent, the unrestricted selector chooses BioViL-T PCA Ridge in 19 of 50 training sets and BioViL PCA Ridge in 16. The best observed extent model is the latter, but the training folds do not identify it consistently. The resulting selector MAE is 0.827, above that pipeline's observed 0.763, while still below bounded OLS 1 at 1.107.

PCA is also selected within training. For BioViL extent, five components with Ridge alpha 0.01 are chosen in 21 of the 50 training sets; ten components with the same alpha are chosen in 13. Reporting these choices is more informative than naming PCA as though it were a single fixed dimensionality chosen from the test results.

Errors on individual images and high scores

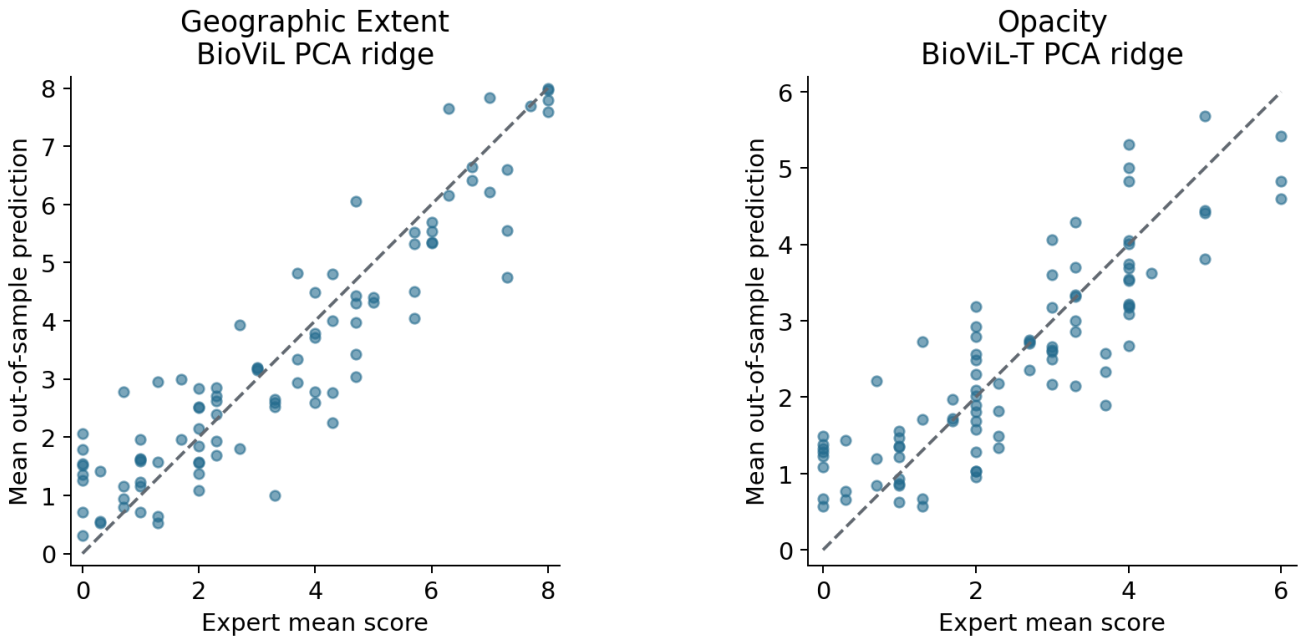


Figure 4. Each point is one of the 94 images. For visualization, its held-out predictions are averaged across the splits in which it was tested. This plot describes repeated prediction behavior and is not a separate test set or a single deployed model.

A smaller overall MAE does not establish that a method handles the cases of greatest concern. Severe-score errors are shown separately because averages can improve while high-score predictions still shrink toward the center. The score labels alone cannot tell us whether such errors would change treatment.

High score cases need a separate reading

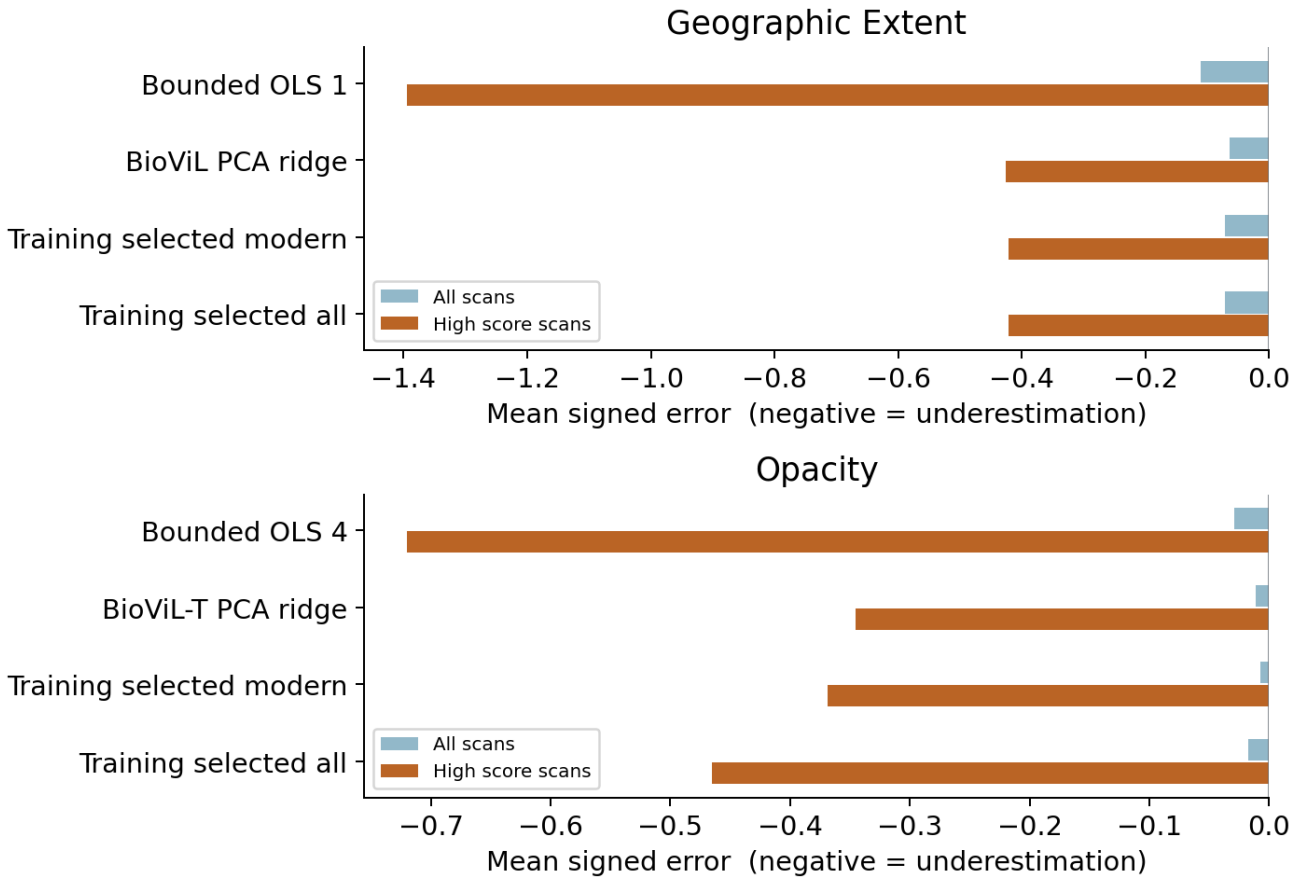


Figure 5. Overall and high-score signed errors. Extent at least 6 and opacity at least 4 are exploratory imaging strata, not validated clinical categories. Negative error indicates underestimation.

The extent result is encouraging: the observed BioViL PCA Ridge candidate lowers high-score MAE from 1.498 to 0.678. The opacity result is more modest in this stratum: BioViL-T PCA Ridge changes high-score MAE from 0.808 to 0.777, despite its larger relative gain in overall error. Training selected all has high-opacity MAE 0.858, worse than the bounded original reference.

Sensitivity analysis and the earlier failure

Pipeline or selection rule	Fivefold extent MAE	Fivefold opacity MAE
Training selected OLS	1.118	0.723
Training selected modern	0.754	0.658
Training selected all	0.754	0.658
Bounded OLS 1	1.097	0.779
BioViL PCA ridge	0.742	0.678
Bounded OLS 4	1.057	0.723
BioViL-T PCA ridge	0.752	0.668

Table 2 pools the 94 unique outer-test predictions in five-fold patient CV. Its training sets are larger, and its partitions differ from the 50 half splits. It should be read as an internal sensitivity analysis, not appended as extra independent observations to the main result.

The benefit of BioViL-based features persists in this sensitivity analysis, but the best opacity configuration changes: the BioViL-T projected SVR becomes more competitive than the PCA Ridge selected from the main leaderboard. This supports a representation-level improvement more clearly than it identifies one universally best regression pipeline.

What the failure teaches about preprocessing

The initial standardized Ridge 1024 pipeline predicted -169.902 for an extent score of 0 in one split. A feature with training SD about 0.00000592 reached a test value of 0.278, giving a standardized value near 46,990. An independent solver reproduced the result. This suggested that scaling was amplifying a feature whose training distribution provided little basis for extrapolation.

DenseNet high dimensional configuration	Extent MAE	Opacity MAE
Ridge 1024	1.419 ± 0.633	0.931 ± 0.515
DenseNet1024 unscaled ridge	1.152 ± 0.138	0.813 ± 0.096
DenseNet1024 filtered ridge	1.146 ± 0.135	0.758 ± 0.086
DenseNet1024 PCA ridge	1.177 ± 0.153	0.816 ± 0.073

The extension tests no feature scaling, a training-only variance filter and PCA. Each also uses score bounds. Those simultaneous changes mean the overall gain is a pipeline gain; it cannot be assigned solely to the variance filter. The original failed result is retained in the initial outputs, and raw expanded predictions allow the remaining numerical behavior to be checked.

Raw OLS predictions with 1, 4 and 18 features match the initial run exactly. The underdetermined OLS 1024 refit has a maximum prediction difference of 0.000808 when changing the BLAS thread count from two to one; refitting with each thread count reproduces the respective stored predictions. This does not affect the low-dimensional clipping controls or the main encoder comparison.

What the results support

Interpretation

The original low-dimensional model is a demanding reference for this dataset: its features already express lung opacity and related abnormalities. A new model has to add information that is not captured by that compact representation, while learning from only a few dozen patients per split. Large embeddings offer more information but also more ways to fit an unstable relationship.

The results favor BioViL-based features, not every newer encoder. All four tested RAD-DINO configurations are worse than the bounded original references in the 50-split comparison. BioViL PCA Ridge also outperforms both tested regressors on the smaller projected BioViL representation for extent. A plausible explanation is that information useful for a continuous extent score is not equally preserved by every pretraining objective or projection. The experiment does not isolate that mechanism, so we treat it as a hypothesis.

For Geographic Extent, the observed modern-encoder candidate is BioViL PCA ridge. Its patient-weighted MAE is 0.787; high-score MAE is 0.678, with signed error -0.426. These describe different aspects of performance and should accompany the headline image MAE.

For Opacity, the observed modern-encoder candidate is BioViL-T PCA ridge. Its patient-weighted MAE is 0.671; high-score MAE is 0.777, with signed error -0.346. These describe different aspects of performance and should accompany the headline image MAE.

Limits that affect the conclusion

All results come from the same 65 patients and subjective expert scores. The dataset is assembled from public sources rather than a consecutive hospital population. Our experiment does not test portable AP images, acquisition shift across hospitals or clinical outcomes. The public pretrained encoders were frozen, so it does not compare every feasible fine-tuning approach.

The declared pretraining sources are documented for the modern encoders, but we have not performed a pixel-level membership audit against all pretraining images. BioViL-T receives no earlier patient image in this experiment, so its temporal capability is not directly tested. The public RAD-DINO checkpoint must be identified separately from the checkpoint used for its original paper results. [5-8]

The expanded design was informed by the initial negative comparison and the diagnosed scaling failure. It is reasonable to use those findings to improve a coursework experiment, but the result remains exploratory. Confirmation would require a new patient cohort, with the chosen representation, preprocessing and regression rule fixed beforehand.

Conclusion for Part II

The checklist supports a limited reading of the original study: it demonstrates that pretrained chest radiograph features can estimate expert severity scores on a small retrospective cohort. The extension provides a broader comparison of that approach, separates simple score bounds from representation changes, and evaluates the cost of model selection. The observed winner should be presented with the paired comparisons and the selection results, rather than as evidence of general clinical superiority.

References and reproducibility

[1] Cohen JP et al. Predicting COVID-19 Pneumonia Severity on Chest X-ray With Deep Learning. Cureus. 2020;12:e9448. https://doi.org/10.7759/cureus.9448

[2] Zhang Louxin. DSA5206 Lecture 9 Assessment of AI Studies in Medicine. 2026. Checklist p. 11; elaboration pp. 14-21. The Type 2 project handout defines Part II.

[3] COVID-19 Image Data Collection. Snapshot 196143d3dd32a2a43ad51d6891d460aaf690d741, 30 March 2020. https://github.com/ieee8023/covid-chestxray-dataset

[4] Original severity implementation and TorchXRayVision. https://github.com/mlmed/covid-severity ; https://github.com/mlmed/torchxrayvision

[5] Boecking B et al. Making the Most of Text Semantics to Improve Biomedical Vision-Language Processing. ECCV 2022. https://arxiv.org/abs/2204.09817

[6] Bannur S et al. Learning to Exploit Temporal Structure for Biomedical Vision-Language Processing. CVPR 2023. https://arxiv.org/abs/2301.04558

[7] Perez-Garcia F et al. Exploring scalable medical image encoders beyond text supervision. Nature Machine Intelligence. 2025;7:119-130. Preprint first posted 2024. https://doi.org/10.1038/s42256-024-00965-w

[8] Microsoft. RAD-DINO public model card. https://huggingface.co/microsoft/rad-dino ; hi-ml inference implementation https://github.com/microsoft/hi-ml

[9] Scikit-learn 1.5.2. Nested versus non-nested cross-validation. https://scikit-learn.org/1.5/auto_examples/model_selection/plot_nested_cross_validation_iris.html

[10] Existing group reproduction, reviewed commit f2127e6b8f0ae84eef618a308adb657be0cb65c0. https://github.com/apothecary11/DSA5206-GROUP-PROJECT

Audit trail

The repository contains the 94 images and their hashes; severity labels with original attribution; fixed features from all four encoders; source revisions and weight checksums; complete outer partitions; metrics and test predictions; selected parameters and warnings; and the original failure diagnostic. results/expanded/verification.json records 3,870 actual outer fits. Selector evaluations reuse candidate predictions and are not counted as extra fitted models.

The main comparison uses 900 initial fits and 2,700 new fits. The five-fold sensitivity analysis adds 270 fits. Inner tuning fits are additional and are not included in those counts. The initial outputs are retained under results; expanded outputs are under results/expanded.

Run code/run_extended_comparison.py for the 50 half splits, repeat with --scheme fivefold for sensitivity, then run code/summarize_extended.py. Cached feature reruns need only the analysis requirements. Re-extracting the encoders additionally needs the image requirements and public weight downloads. Exact installed versions are recorded in results/expanded/environment.json.

中文实验说明

老师要求我们用课堂 checklist 评价原论文，并选择替代方法做比较。这次修订保留了第 9 讲的 12 项评价，把实验从“更换回归器”扩展到“更换胸片表示”。原 DenseNet 之外，实际提取了 BioViL、BioViL-T 和 RAD-DINO 的特征。各模型使用自己的公开推理预处理，参数保持冻结。

病变范围中，BioViL PCA ridge 的 MAE 为 0.763，相对于相同分数约束下的原方法 1.107，误差降低 31.1%。

浑浊程度中，BioViL-T PCA ridge 的 MAE 为 0.662，相对于相同分数约束下的原方法 0.745，误差降低 11.2%。

这些是本次重新计算的数值。图中的 Published OLS 表示按原论文方法重新运行的基线，不是从论文表格复制的数据。比较时也给原方法加上同样的分数范围约束，避免把简单截断带来的收益算成新模型的贡献。

为什么要报告两种结果

我们可以在完整排行榜中找出测试误差最低的方法，但尝试的方法越多，越容易挑中一个刚好适合这批患者的方法。为此，报告同时给出仅利用训练集内部验证来选模型的结果。如果排行榜冠军不错，而训练内选模的结果一般，说明当前样本还不足以稳定识别冠军，不能把最低的那个数字当作确定结论。

本次新增 27 组配置，连同原来的 9 组共有 36 组实际模型配置。50 次划分的主实验运行 3,600 次外层拟合，五折敏感性分析再运行 270 次，合计 3,870 次。三种选模程序复用候选模型的预测，不重复计算成新的拟合。

如何阅读五折结果

五折患者分组验证中，每位患者只进入一次测试折，训练患者比原来的半数划分更多。它可以帮助判断结果对训练样本数量和划分方式是否敏感。不过，两种实验仍然使用同一批患者，不能把五折结果当作外部验证。

这次没有完成的事情

我们没有重新训练这些大型编码器，也没有复现它们原论文中的全部任务。BioViL-T 在这里仅接收当前胸片，没有输入历史影像。数据仍只有 65 位患者，因此无法回答跨医院泛化或临床应用效果。报告中的改善应限定为本数据、本设置下的影像评分误差变化。

中文讨论与汇报思路

原方法为什么值得保留

原论文的低维模型并不只是“年代较早的简单模型”。它先利用大量胸片上预训练的网络，再只保留与肺部浑浊等相关的少量输出。这些特征与严重程度评分接近，而需要在小样本上估计的参数又少，因此是有针对性的强基线。与它比较，比只挑一个表现较差的高维回归器更有说服力。

新模型的收益应如何解释

较新的编码器可能保留更多影像信息，但它们的预训练目标并不等同于这两个严重程度分数。我们需要检查改善是否同时出现在患者平均误差、高分病例和不同划分方式中。若只改善总体 MAE，而高分病例依然被低估，结论应当保留这一差别，不能写成“重症判断更加准确”。

高维 Ridge 的问题给了什么线索

第一次实验中的极端负分与一个近乎常量的特征有关。训练集标准差很小，标准化就会把测试时的偏移放大。换一个求解器仍得到同样结果，说明问题不是单纯的数值求解误差。这次分别加入无标准化、低方差过滤和 PCA，目的是检查模型表现到底受表示维度还是受处理方式影响。由于新增流程也使用了分数约束，不能把全部改善只归因于某一个步骤。

checklist 怎样接到实验上

数据收集和数据质量的评价说明为什么结果的适用范围有限；模型选择和数据划分的评价决定我们怎样设计公平比较；指标、解释性和泛化问题则决定我们怎样看待最低 MAE。这样，checklist 就不是一张与实验分离的打分表，而是实验设计和结论范围的依据。

汇报时的表述

可以先解释原论文把胸片映射到两个专家评分的思路，再指出最值得补充的是表示模型、预处理稳定性和选模偏差。展示主要结果时，同时放原方法、相同约束下的原方法、表现较好的新方法，以及训练内选模程序。最后用高分病例误差和五折结果交代改善的范围。

最有根据的后续工作，是在新医院患者上检验已经选定的完整流程。继续在这 65 位患者上增加模型，可以帮助理解问题，但不能代替新数据对结论的检验。